



How could observed sowing dates contribute to maize potential yield under climate change in Northeast China based on APSIM model

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ABSTRACT

The Northeast China (NEC) accounts for more than 30% of the national total maize planted area (grain yield). Adjusting the sowing dates has been considered an effective measure to adapt to climate change, but there was little evidence that how well the producers had done. In this study, we used phenology observations at 67 stations from 1981 to 2014 to detect trends in actual sowing dates, then the agricultural production systems simulator, APSIM-Maize model, was used to assess the effects of changes in observed sowing dates on maize phenology and yields. During the past 34-year period, the actual maize sowing dates show a delaying tendency, at a rate of 1–6 days per decade, but there are significant fluctuations among years. For per day delay in the sowing dates, the whole growing season was shortened by 0.1%. Delaying sowing dates reduced the solar radiation interception during the vegetative period as well as the thermal time during the reproductive period. As a result, the overall maize potential yield was negatively affected in NEC; for per day delay in the sowing dates, the potential yield was decreased by 0.6%. By contrast, advancing sowing dates in some years led to increases in both the solar radiation interception during the vegetative period and the thermal time during the reproductive period. However, these increases showed various effects on the maize potential yield across different parts of the study region. We detected a positive effect of advancing sowing dates on maize potential yield in the high latitudes, at a rate of up to 1.6%. By contrast, in the low latitudes, the negative effect of advancing sowing dates on maize potential yield was dominant, at a rate of up to 2.7%.

1. Introduction

Maize is not only an important food crop but also an important industrial and feed material (Shi et al., 2001). China is the second largest maize producer in the world; and Northeast China (NEC) is a main maize-growing region in China. Since 2013, China's maize planted area (grain yield) has accounted for more than 30% of the national total food crop planted area (grain yield). Regional maize planted area (grain yield) in NEC accounts for more than 30% of the national maize planted area (grain yield) (National Bureau of Statistics of China, 2021). The stability of high maize grain yield in NEC is critical to national maize production and, hence, food security. The Fifth Assessment Report of the Intergovernmental Panel on Climate Change (IPCC) pointed out that the global average temperature has risen by 0.85 °C over the period 1880 to 2012, and the forecast showed that the global temperature change will likely to exceed 1.5 °C relative to 1850 to 1900 in the end of the 21st

century (IPCC, 2013). NEC is in relatively high latitudes, and the regional temperature has been significantly affected by this global warming. On a regional scale, the starting date of the potential growing period with daily average air temperature above 10 °C advanced, and the ending date of the potential growing period delayed, thus, length of the potential growing period of maize lengthened. During the lengthened growing season for maize, thermal time has been increasing while solar radiation interception and precipitation both have been decreasing (Yang et al., 2011).

Changes in growing season temperature, precipitation, and solar radiation interception all affect maize production. In particular, rising temperatures could accelerate the growth rate of crop plants and shorten the length of the growth period for maize, which is not conducive to the accumulation of biomass or the formation of yield (Lobell and Field, 2008; Shao et al., 2021). In NEC, without considering adaptive measures, climate change caused a 22–26% reduction in regional maize

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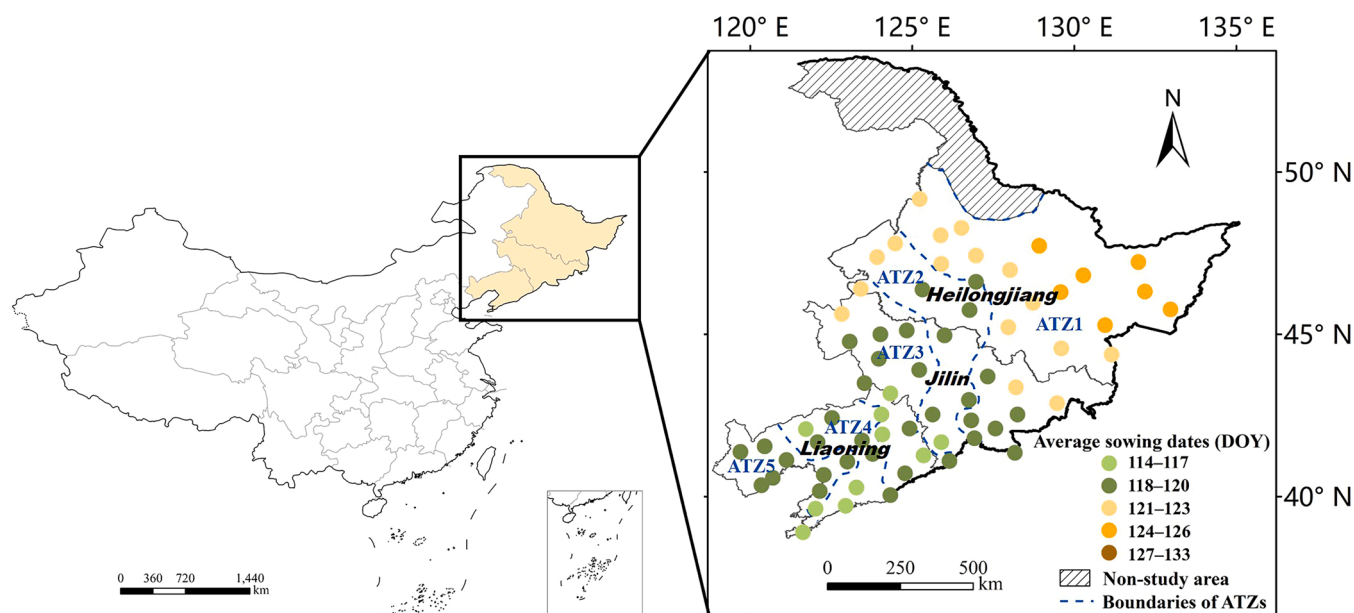


Fig. 1. Average sowing dates at the 67 agrometeorological stations from 1981 to 2014 in the five accumulated temperature zones (from east to west: ATZ1, ATZ2, ATZ3, ATZ4, and ATZ5) for maize in Northeast China. Note: Provinces are named in italics and their boundaries are shown in gray lines. Sowing dates are expressed in day of year (DOY).

potential yield from 1961 to 2010 (Lv et al., 2013; Zhao, 2015; Liu et al., 2018). However, in the high latitudes of NEC where climate was usually cool, the warming trend significantly lengthened the potential growing period for maize. When properly adapting agricultural management practices, the natural resources in the potential growing season could be effectively utilized to improve yield production. Possible adaptive measures include adjusting the system of planting (Yang et al., 2015), changing cultivars (Liu et al., 2013; Su et al., 2021), optimizing sowing dates (Zhang et al., 2018; Bai et al., 2020), and modifying water and fertilizer applications (Liu et al., 2016b).

Previous studies of sowing dates effects on maize were mostly based on field experiments and crop modeling. In general, delaying sowing dates could shorten the growth period of maize plants, which reduced the radiation use efficiency and hence the photosynthesis products formation. By contrast, appropriately advancing sowing dates could lengthen the growth period of maize plants, which provided favorable environmental conditions for the photosynthetic performance at the grain filling stage and leads to increases in the use efficiency of photosynthetic active radiation (PAR) and the photosynthetic products formation (Liu et al., 2009; Xiao et al., 2014; Yu et al., 2013; Zhang et al., 2018). Delaying sowing dates could decrease the hundred kernel weight, ear number, ear length, ear thickness, row number, and kernel number per row in maize; for per day delay in the sowing dates, maize grain yield would decrease by 0.7–1.7% (Shi and Xu, 2010; Cao et al., 2013).

In NEC, previous studies merely focused on the temperature to determine the initial sowing dates and then certain backward steps could be set to explore the optimal sowing dates based on the field experiments or the crop model (Bai et al., 2020; Lei et al., 2014; Qian and Guo, 2013). However, the actual sowing dates are also affected by other non-temperature factors, such as precipitation, irrigation facility, labor availability, field condition, and so on. Regional-scale studies of actual sowing dates effect on maize in recent decades have rarely been reported. Agricultural Production Systems sIMulator (APSIM) has been commonly used in impacts assessment of climate change and farm management-decision making processes around the world (Liu et al., 2016a; Yang et al., 2018; Archontoulis et al., 2020). Therefore, we used the APSIM model to determine the contributions of observed sowing dates to the maize potential yield, which would help to increase maize yields sustainably in NEC. The objectives of this study are (i) to identify

overall trends and interannual fluctuations in observed sowing dates of maize over the past 34 years across different parts of the study region; (ii) to assess the effects of observed sowing dates on maize yield and identify reasons, including the duration of maize growing periods and allocation of climatic resources during vegetative period, reproductive period, and whole growing season over the past 34 years in NEC.

2. Materials and methods

2.1. Study region

The study region is located in NEC, comprised of Heilongjiang (north), Jilin (center), and Liaoning (south) Provinces. The heat resources in the northern part of Heilongjiang Province are difficult to meet the required for maize. Therefore, we calculated the annual growing degree day (GDD) from 1981 to 2014, and took the region that above 2100 °C·d as the study region (Fig. 1):

$$GDD = \sum_{i=1}^{365} T_i, \quad T_i = \begin{cases} T_i & T_i > T_b \\ 0 & T_i < T_b \end{cases} \quad (1)$$

Where T_i is the average daily temperature (°C) and T_b is a crop-specific baseline temperature. For maize, a T_b of 10 °C was used (Wang, 2004), and 2100 °C·d is the GDD threshold for planting maize in NEC (Yang, 2007).

In order to up-scale location-specific assessment of sowing dates and its impact on the maize potential yield, the study region was further divided into 5 accumulated temperature zones (ATZs) by dividing the GDD into 5 equal parts (Liu et al., 2016b). The study region has a temperate monsoon climate, with a warm moist summer and abundant solar radiation interception throughout the year; the soil is fertile and rainfed maize is dominant. During the potential growing period of maize, the average temperature was around 17.0–22.0 °C, and total precipitation was about 450–550 mm. Overall, growing season precipitation was relatively low in the western part of the study region, but it was still enough to meet the basic needs of maize plants.

2.2. Data

We obtained daily weather data from 1981 to 2014 for 67 weather stations in the study region from the China Meteorological Data Network (<https://data.cma.cn>), including average, maximum, and minimum temperatures, precipitation, sunshine hours. We acquired the concurrent routine observations for maize at 67 agrometeorological stations in the study region, including variety, management practices (i.e., plant density, plant depth, row spacing, and irrigation and fertilizer applications), phenological events (i.e., sowing, flowering, and physiological maturity dates), and yield. Aside from the soil data provided by the agrometeorological stations, we also referred to the “Second Soil Census Farmland Fertility Database” to derive the soil bulk density, wilting point, field capacity, and saturated water content in the 0–100 cm soil layer. In the analysis, we divided the whole growing season for maize into two periods: vegetative period (emergence to flowering, VP) and reproductive period (flowering–physiological maturity, RP).

2.3. Crop modeling

Crop potential productivity is the maximum possible yield per unit field area (Hou and Liu, 1985), or the maximum theoretical yield level that could be achieved given the climate, soil, and agricultural environment (Wu and Han, 1991). The photosynthesis-temperature potential productivity is the upper limit of theoretical yield determined by solar radiation interception and temperature, with suitable soil, abundant water resource, and desirable agricultural management practices (Li et al., 1988). Our previous study used the observations from the agrometeorological stations to calibrate the APSIM-Maize model for representative varieties in the 1980 s, the 1990 s and the 2000 s for each ATZ respectively, and the result indicated that APSIM-Maize performed well in simulating growth and yield for maize in NEC. For the days of flowering and maturity, the coefficient of determination (R^2) values was 0.83–0.89 and 0.77–0.98 and the D values was 0.90–0.96 and 0.90–0.97, respectively. In addition, simulated yields compared well with observed yields (R^2 0.73–0.94; D -value 0.80–0.93); more details are available in Liu et al. (2012, 2016b) and Bai et al. (2020). In this study, we rely on the previous model validation work for model performance. In the validated model, we simulated station-level maize potential yield from 1981 to 2014 in the study region. The technology and management consist of sowing depth (5 cm), row spacing (60 cm), and plant density (90,000 plants·ha⁻¹). Nitrogen fertilizer applications were taken as nonlimiting, and irrigation would be carried out automatically when the soil water content falls below 80% of the field capacity to ensure that plants were grown without water or nitrogen stresses. In the sowing dates effect analysis, we compared the maize potential yield under two scenarios: a) sowing dates were the date actually observed by the agrometeorological stations, which differed year by year; and b) sowing dates were fixed to be the average sowing dates observed during the period of 1981–1985.

2.4. Data analysis

We calculated the difference for sowing dates, duration of maize growing periods, and allocation of climatic resources during vegetative period, reproductive period between the two scenarios (Eq. (2)) so as to explore the fluctuation of sowing dates among years and the changes of growth period and resource allocation caused by it.

$$D = X_a - X_f \quad (2)$$

Where D is the value of changes, X_a is the sowing dates, duration of growing period, and agri-climatic resources under the actual sowing dates scenario; X_f is the sowing dates, duration of growing period, and agri-climatic resources under the fixed sowing dates scenario. $D > 0$ ($D < 0$) indicates delayed (advanced) sowing dates, extended (shortened)

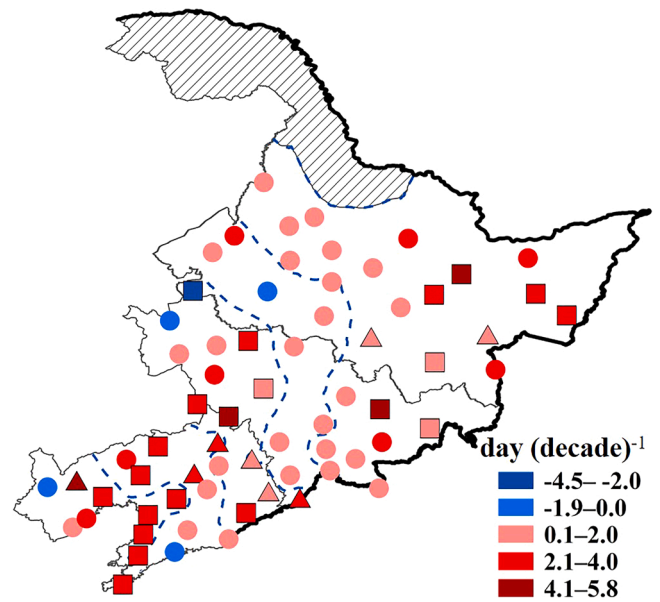


Fig. 2. Geographic distribution of the decadal trends in actual maize sowing dates during the period of 1981–2014 for the 67 study stations in Northeast China. Note: a square symbol indicates a statistically significant trend at $p < 0.01$, a triangle symbol indicates statistically significant trend at $p < 0.05$, and a circle symbol indicates statistically not significant trend ($p > 0.05$).

growing period, and increased (decreased) resources.

We used the relative rate of change in simulated maize potential yield under the two sowing dates scenarios to compute the yield contribution rate (Eq. (3)). In addition, we conducted linear regression analysis between the sowing dates and years, between sowing dates and potential yield and evaluated the significance level of the linear regression slope (Eq. (4)) with Student's t -test at 95% and 99% confidence intervals (CIs).

$$R_c = \frac{Y_a - Y_f}{Y_f} \times 100\% \quad (3)$$

Where R_c is the yield contribution rate; Y_a is the maize potential yield under the actual sowing dates scenario; and Y_f is the maize potential yield under the fixed sowing dates scenario. $R_c > 0$ ($R_c < 0$) indicates a positive (negative) effect of changing sowing dates on the yield.

$$Y_i = a + bX_i \quad (4)$$

Where a is the constant of regression; b is the slope of regression. For the analysis of the trends of sowing dates, Y_i is the sowing date and X_i is the year; For the analysis of the impacts of sowing dates on maize potential yield, Y_i is the changes in yield and X_i is the changes in sowing dates.

3. Results

3.1. Observed maize sowing dates

From 1981–2014, 62 of the study stations showed delaying trends in maize sowing dates. Among these, 28 of the study stations showed statistically significant ($p < 0.05$) delaying trends, with rates of 1.3–5.8 days per decade, and 1 station showed statistically significant ($p < 0.01$) advancing trends. At the stations that showed statistically significant delaying trends, the amplitude was relatively higher in central Liaoning and eastern Heilongjiang (more than 2.0 days per decade) and relatively lower in the central area of the study region (fewer than 2.0 days per decade) (Fig. 2).

In the following sections, we selected these 28 stations with statistically significant delaying trend for the fluctuation analysis of sowing

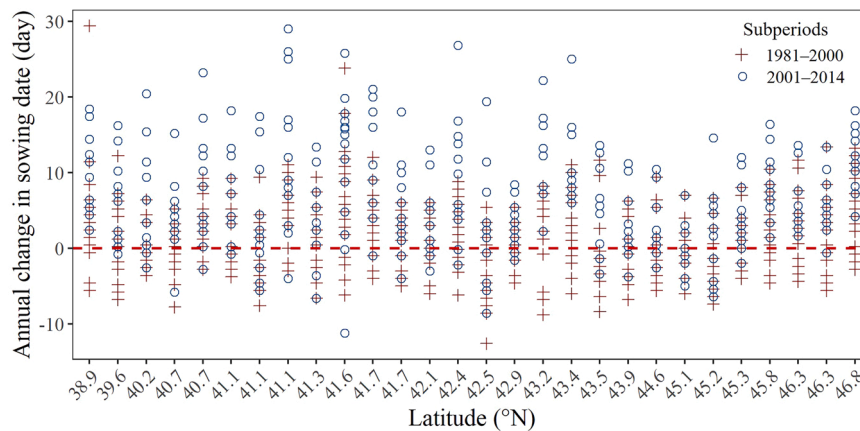


Fig. 3. Annual change in the actual maize sowing dates during the two subperiods of 1981–2000 and 2001–2014 compared to the average value in 1981–1985 for the 28 selected study stations.

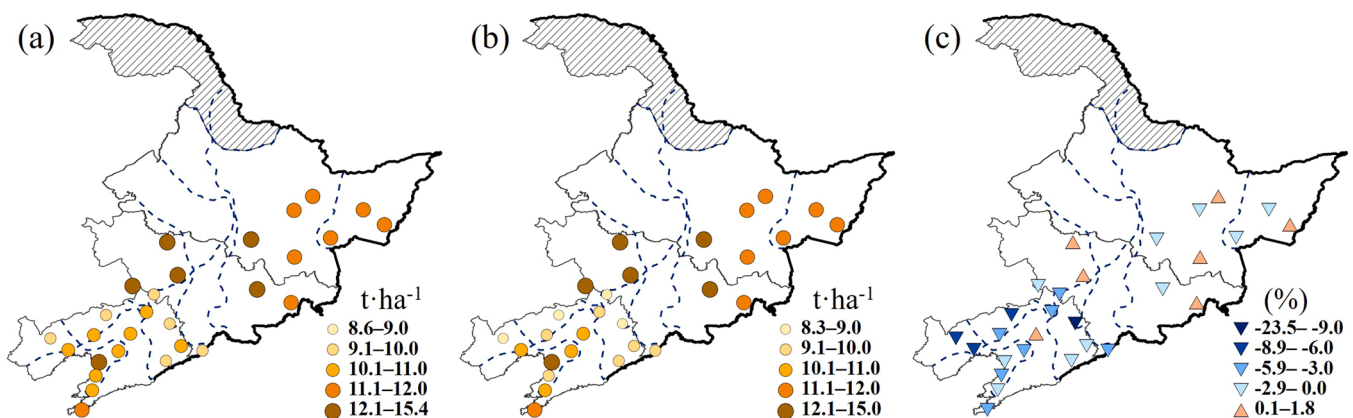


Fig. 4. Geographical distribution of the mean maize potential yield under (a) the fixed sowing dates scenario and (b) the actual sowing dates scenario during the period of 1981–2014 at the 28 selected study stations. (c) The change rate of mean maize potential yield under the actual sowing dates scenario vs. under the fixed sowing dates scenario.

dates and effects of sowing dates on potential yield, growing period, allocation of climatic resources of maize. Compared to the baseline from 1981 to 1985, the actual maize sowing dates at these 28 stations mostly changed by -5 (advance) to 10 days (delay) during the entire study period of 1981–2014. It is worth noting that the annual changes in the actual maize sowing dates fluctuate more at the stations in low latitudes (more than 20 days in some years) than at the stations in high latitudes (Fig. 3). This points that compared with the early 1980 s, the actual inter-annual sowing dates were both advanced and delayed, although the long-term series shows a significant trend of delay.

3.2. Effects of sowing dates on potential yield of maize

During the period of 1981–2014, mean maize potential yield at the 28 selected stations ranged from 8.6 to 15.4 t ha^{-1} (from 8.3 to 15.0 t ha^{-1}) under the fixed (actual) sowing dates scenario. Mean maize potential yield showed similar spatial patterns under the two sowing dates scenarios, with high values (above 10.0 t ha^{-1}) scattered in ATZ1 and ATZ3. At most of the stations in ATZ3, ATZ4, ATZ5, mean maize potential yield was about 3.0 – 6.0% lower under the actual sowing dates scenario than under the fixed sowing dates scenario. At few stations in ATZ1, mean maize potential yield was about 0.1 – 1.8% higher under the actual sowing dates scenario than under the fixed sowing dates scenario (Fig. 4). Although the actual sowing dates drastically fluctuated compared to the baseline, there was no substantial difference between mean maize potential yield under the two sowing dates scenarios. This

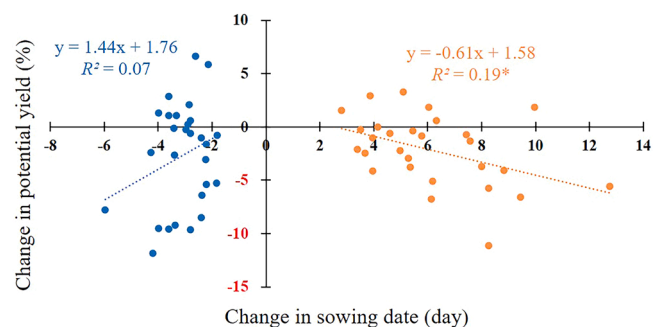


Fig. 5. Linear regression between change in maize potential yield and change in sowing dates from 1981 to 2014 at the 28 selected stations. Note: A blue symbol indicates advanced sowing dates and an orange symbol indicates delayed sowing dates; the same definitions also apply in Fig. 7(b).

was because the composite value masked the effects of sowing dates advance and delay on maize potential yield.

Furthermore, we divided the annual change in sowing dates at each of the selected stations into two types (delayed and advanced) to analyze the corresponding effects on maize potential yield. Overall, delayed sowing dates showed a negative effect on maize potential yield at the 28 selected stations; for per day delay in sowing dates, maize potential yield decreased by 0.6% on average. Advanced sowing dates showed a

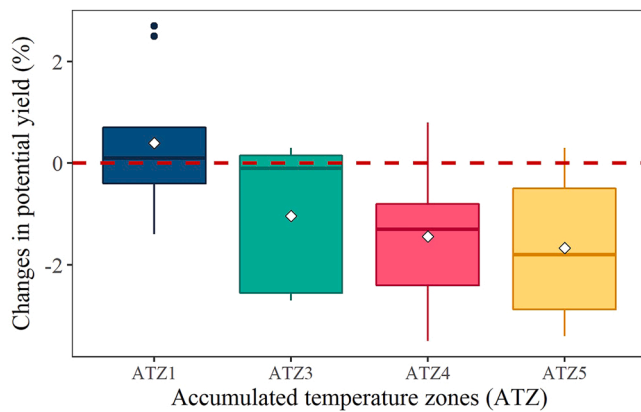


Fig. 6. Box plot of the corresponding changes in maize potential yield for per day advance in sowing dates at the selected stations in the four accumulated temperature zones of ATZ1, ATZ3, ATZ4, and ATZ5 during the period of 1981–2014. Note: The dashed red line indicates zero-change in the potential yield.

positive effect on maize potential yield at most of the stations in ATZ1 (high latitudes) but a negative effect on maize potential yield at most of the stations in ATZ3, ATZ4, and ATZ5 (low latitudes). For per day advance in sowing dates, maize potential yield was decreased by 1.0%, 1.5%, and 1.7% on average in ATZ3, ATZ4, and ATZ5, respectively (Fig. 5). Among the four accumulated temperature zones, for per day advance in sowing dates, the amplitude of decrease in maize potential yield was higher in relatively low latitudes than in relatively high latitudes (Fig. 6). This could be related to the difference in climatic resources between the high and low latitudes.

3.3. Effects of sowing dates on growing period of maize

From 1981–2014, advanced sowing dates lengthened the vegetative period for maize by 0.5–2.4 days at the selected study stations. By contrast, advanced sowing dates shortened the reproductive period, and the shortening effect was greater in high latitudes ATZ1 (by 0.6 days on average) than in low latitudes ATZ3, ATZ4, and ATZ5 (by 0.4, 0.3, and 0.1 days, respectively). Collectively, advanced sowing dates lengthened the growing season for maize by 0.8, 0.7, 1.3, and 1.1 days in ATZ1, ATZ3, ATZ4, and ATZ5, respectively. On the other hand, delayed sowing dates shortened the vegetative period of maize by 1.5, 1.1, 2.0, and 2.4 days in ATZ1, ATZ3, ATZ4, and ATZ5, respectively. Meanwhile, delayed sowing dates lengthened the reproductive period by 0.3–2.6 days. As a result, delayed sowing dates shortened the growing season for maize more in ATZ5 (by 1.5 days) than in ATZ1, ATZ3, and ATZ4 (by 0.3, 0.8, and 0.9 days, respectively) (Fig. 7a). In the entire study region, for per day advance in sowing dates, the length of the vegetative period increased by 0.45% and the length of the growing season for maize increased by 0.28%. For per day delay in sowing dates, the length of the vegetative period decreased by 0.36%, the length of the reproductive period increased by 0.30%, and the length of the growing season for maize decreased by 0.09% (Fig. 7b).

3.4. Effects of sowing dates on allocation of climatic resources during the maize growing season

We further divided the years of each sowing dates type into positive contribution or negative contribution according to the $R_c > 0$ or $R_c < 0$, analyzed the effects of sowing dates on the allocation of climate resources to explain the impact path it on yield. Advanced sowing dates increased the total solar radiation interception (mostly by up to 40 MJ

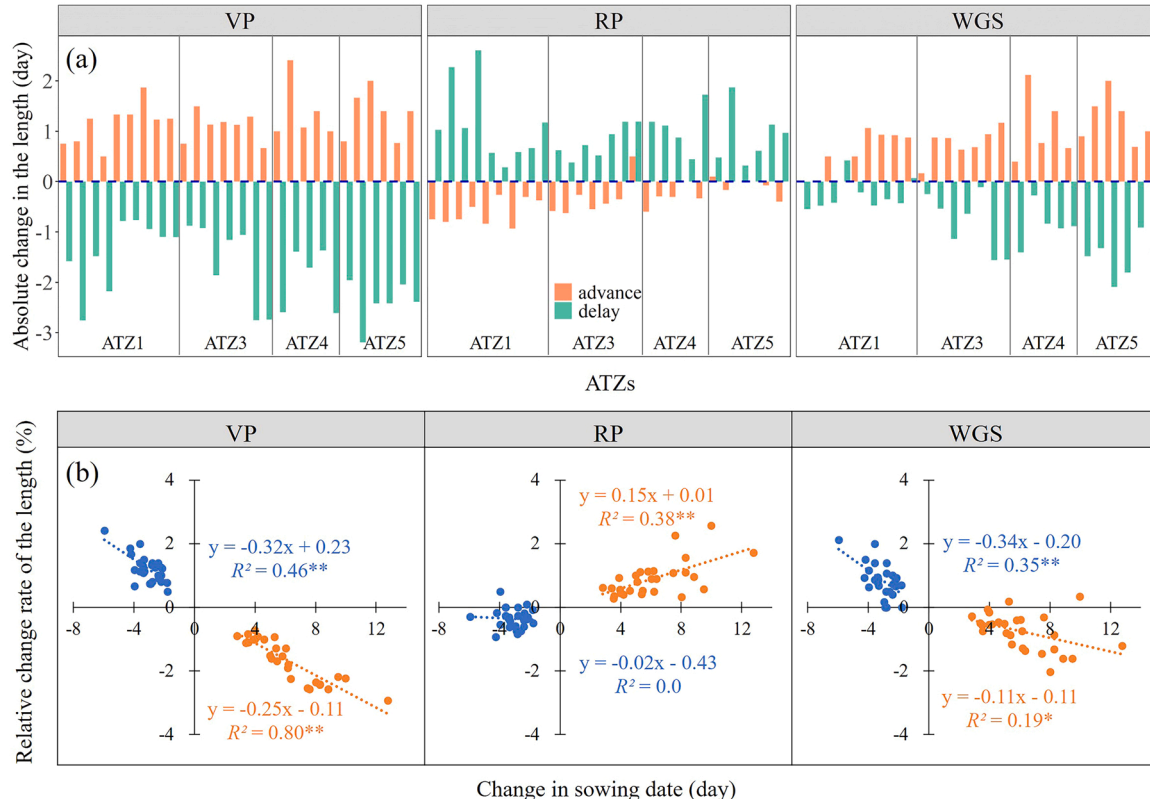


Fig. 7. (a) Histograms of the absolute change in the length of the growth period for maize (i.e., vegetative period, reproductive period, and whole growing season) corresponding to the advance (orange) and delay (green) in sowing dates from 1981 to 2014 at the selected study stations in ATZ1, ATZ3, ATZ4, and ATZ5. (b) Linear regression between the relative change rate of the length of the growth period for maize and the change in sowing dates during the period of 1981–2014 at the selected stations in the study region.



Fig. 8. Box plot of the change in climate resources (i.e., total solar radiation interception, as well as average, maximum, and minimum temperatures) during the vegetative period, reproductive period, and whole growing season for maize due to changes in sowing dates during the period of 1981–2014 at the selected stations in ATZ1, ATZ3, ATZ4, and ATZ5. (a) Advanced sowing dates. (b) Delayed sowing dates.

m^{-2}) but decreased the average, maximum, and minimum temperatures (mostly by $0.1\text{--}0.6\text{ }^{\circ}\text{C}$) during the vegetative period. Advanced sowing dates changed the total solar radiation interception by $-20\text{--}15\text{ MJ m}^{-2}$ and increased the temperatures by up to $0.3\text{ }^{\circ}\text{C}$ during the reproductive period. Overall, advanced sowing dates changed the total solar radiation interception during the growing season for maize by the same amount as during the vegetative period; advanced sowing dates decreased the temperatures during the growing season mostly by $0.1\text{--}0.2\text{ }^{\circ}\text{C}$ (Fig. 8(a)). By contrast, delayed sowing dates decreased the total solar radiation interception (by up to 40 MJ m^{-2}) and increased the temperatures (mostly by up to $0.9\text{ }^{\circ}\text{C}$) during the vegetative period; delayed sowing dates decreased the temperatures during the reproductive period (mostly by up to $0.5\text{ }^{\circ}\text{C}$) (Fig. 8(b)).

Advanced sowing dates showed mixed effects on the temperatures during the vegetative period, and the amplitude of temperature decrease was lower in the positive-correlation years than in the negative-

correlation years. Advanced sowing dates showed a greater effect on the temperatures during the reproductive period in ATZ1 and ATZ3 (relatively higher latitudes) than in ATZ4 and ATZ5 (relatively lower latitudes). Due to the advance in sowing dates, the maximum (minimum) temperature during the reproductive period in ATZ1, ATZ3, ATZ4, and ATZ5 changed by $0.18, 0.10, 0.07,$ and $0.05\text{ }^{\circ}\text{C}$ ($0.20, 0.15, 0.16$ and $0.08\text{ }^{\circ}\text{C}$), respectively, and the average temperature during the reproductive period in ATZ1, ATZ3, ATZ4, and ATZ5 changed by $-0.20, 0.12, 0.10,$ and $0.06\text{ }^{\circ}\text{C}$, respectively (Fig. 8(a)).

Delayed sowing dates mostly showed an increasing effect on the total solar radiation interception during the reproductive period in the positive-correlation years, by up to 20 MJ m^{-2} . By contrast, delayed sowing dates changed the total solar radiation interception during the reproductive period by $-10\text{--}10\text{ MJ m}^{-2}$ in the negative-correlation years, although the decreasing effect was dominant. The amplitudes of temperature decrease during the reproductive period due to the delay in

sowing dates was higher in the positive-correlation years than in the negative-correlations years (Fig. 8(b)).

4. Discussion

Previous studies used both field experiments and crop modeling to optimize the sowing dates for maize in NEC. In addition to being affected by weather, field experiments were also affected by labor, seed quality, farming equipment, etc. Therefore, field experiments alone are not sufficiently reliable for long-term effect studies. In previous studies that used crop modeling to determine the optimal sowing dates, sowing dates were hypothetical and did not represent the actual sowing dates in maize production. In this study, we combined the advantages of both field experiments and crop modeling to analyze the interannual fluctuation and long-term trends of actual sowing dates, and its effect on maize in NEC. Our study results could be used to assist the local agricultural community to optimize sowing dates in order to achieve high maize yields.

Advanced sowing dates showed a positive effect on the maize potential yield, especially in the high latitudes of the study region. By contrast, delayed sowing dates showed a negative effect on the maize potential yield in most of the study years and a positive effect in a few of the study years. These results are consistent with some of the previous related studies (Xiao et al., 2014; Yu et al., 2013). However, the actual sowing dates under long time series still shows delayed trend mainly, we consider that there are three main reasons. First of all, the interannual fluctuations suggests that in some years the sowing dates were actually advanced; secondly, farmers usually determine a sowing date within a window period based on experience after considering rainfall, soil moisture, machinery and labor and so on, instead of being able to determine which day to sow can get the optimum yield at harvest. Thirdly, according to the previous related studies, the main effective measures to adapt maize production to climate change include switching varieties, modifying irrigation and fertilizer applications, changing plant density, adjusting sowing dates, etc. Lv et al. (2013) reported that changing variety could increase maize yield by 46.1–79.0% in Jilin province in China. Mueller et al. (2012) pointed out that the yield of most crops could be increased by 45–75% through improving field fertilization and irrigation management. Cardwell (1982) found that the maize plant density in the United States increased by 1000 plants ha⁻¹ per year from the 1930s to the 1980s, and the increase in plant density contributed about 21.0% to the yield increase. Bai et al. (2020) concluded that optimizing sowing dates for maize could contribute to a 2.4–9.7% increase in the grain yield in NEC. Farmers may therefore consider several adaptative measures simultaneously.

Hence, the influence of rainfall, soil moisture, labor and other factors should be taken into the future research, and we strongly urge the local maize community to weigh the immediate pros and potential cons of delaying sowing dates before making decisions.

5. Conclusions

The actual sowing dates for maize in NEC showed a delayed trend during the past 34 years, at a rate of 1.3–5.8 days per decade. However, compared with the early 1980s, the sowing dates fluctuated greatly among years with a range of – 5–10 d.

Advanced sowing dates showed the opposite effects on maize potential yield among ATZs, a positive effect in high latitudes (ATZ1) but a negative effect in low latitudes (ATZ3, ATZ4, and ATZ5). Because although advanced sowing dates shortened the reproductive period, the increase in total solar radiation interception and temperature in ATZ1 was greater than that in low latitudes. Delayed sowing dates showed an overall negative effect on maize potential yield in the study region. And this was because a) the shortening in the vegetative period resulted in weak plants; b) the decreases in temperatures during the reproductive period reduced the grain filling efficiency.

CRedit authorship contribution statement

Z.J. Liu, X.G. Yang and G.X. Zhu contributed to the study conception and design; material preparation, data collection and analysis were performed by G.X. Zhu, Z.J. Liu, S.L. Qiao, Z.T. Zhang, Q.W. Huang and Z.E. Su; the first draft of the manuscript was written by G.X. Zhu; Z.J. Liu and S.L. Qiao commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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